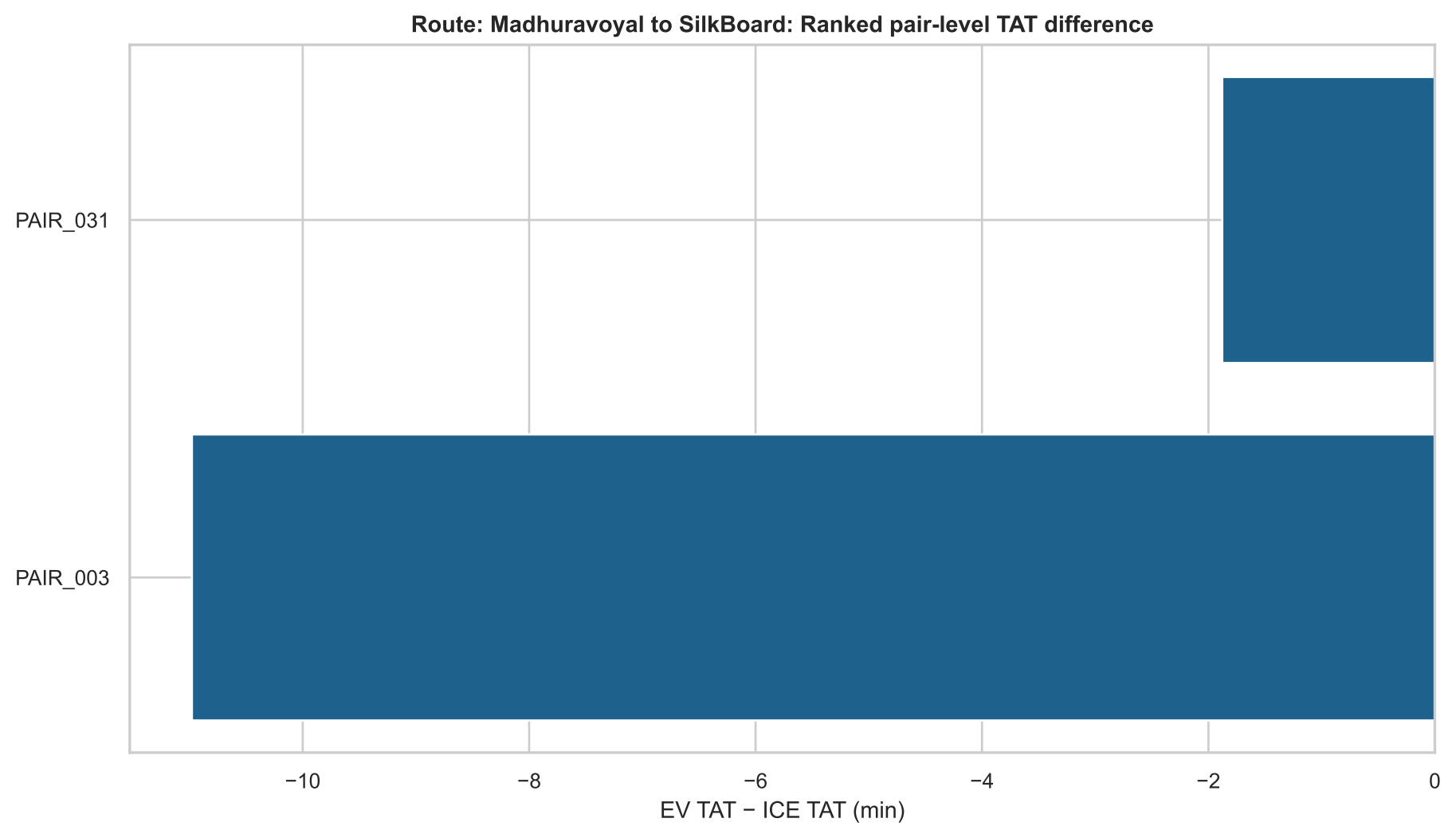
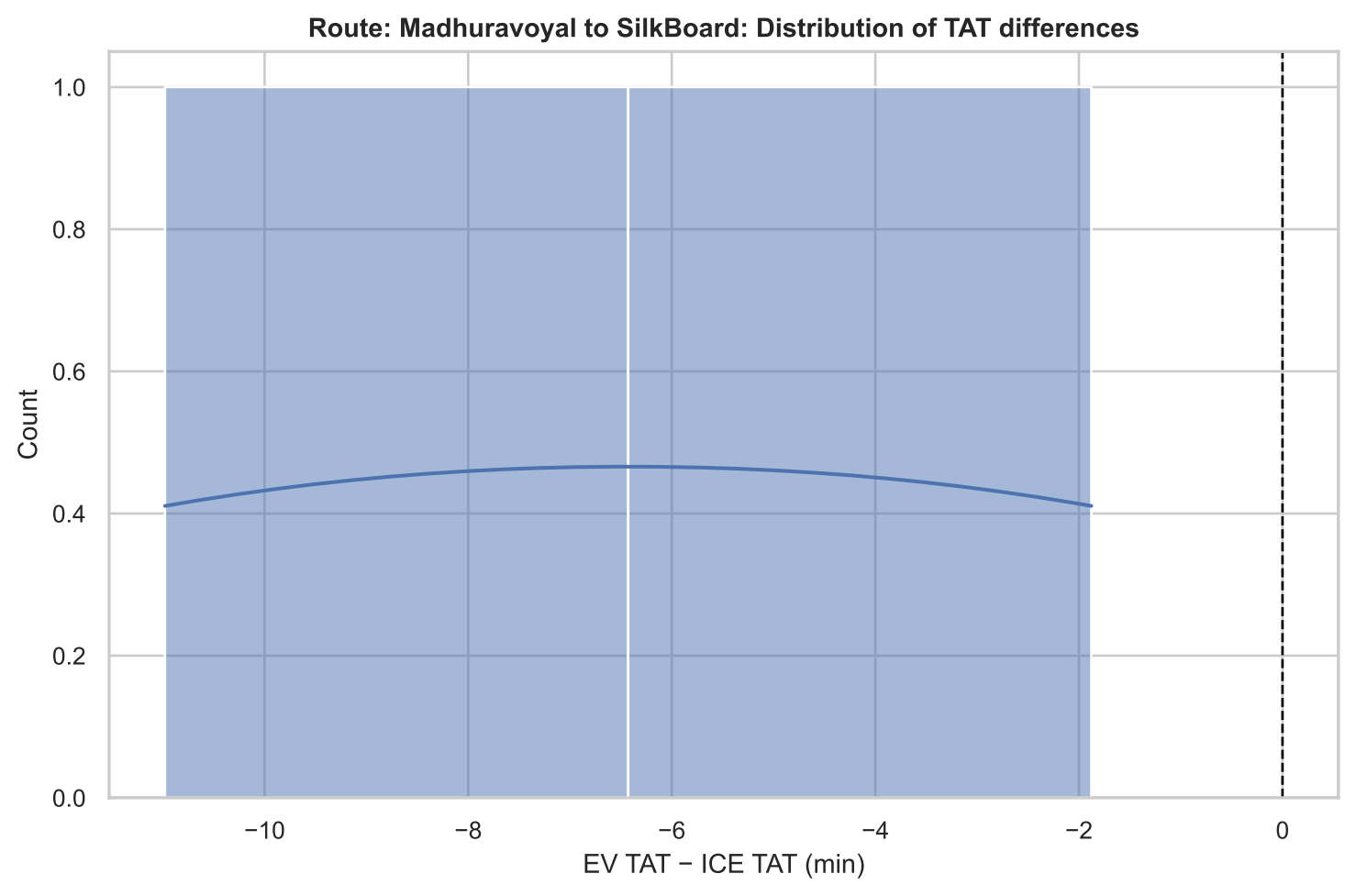
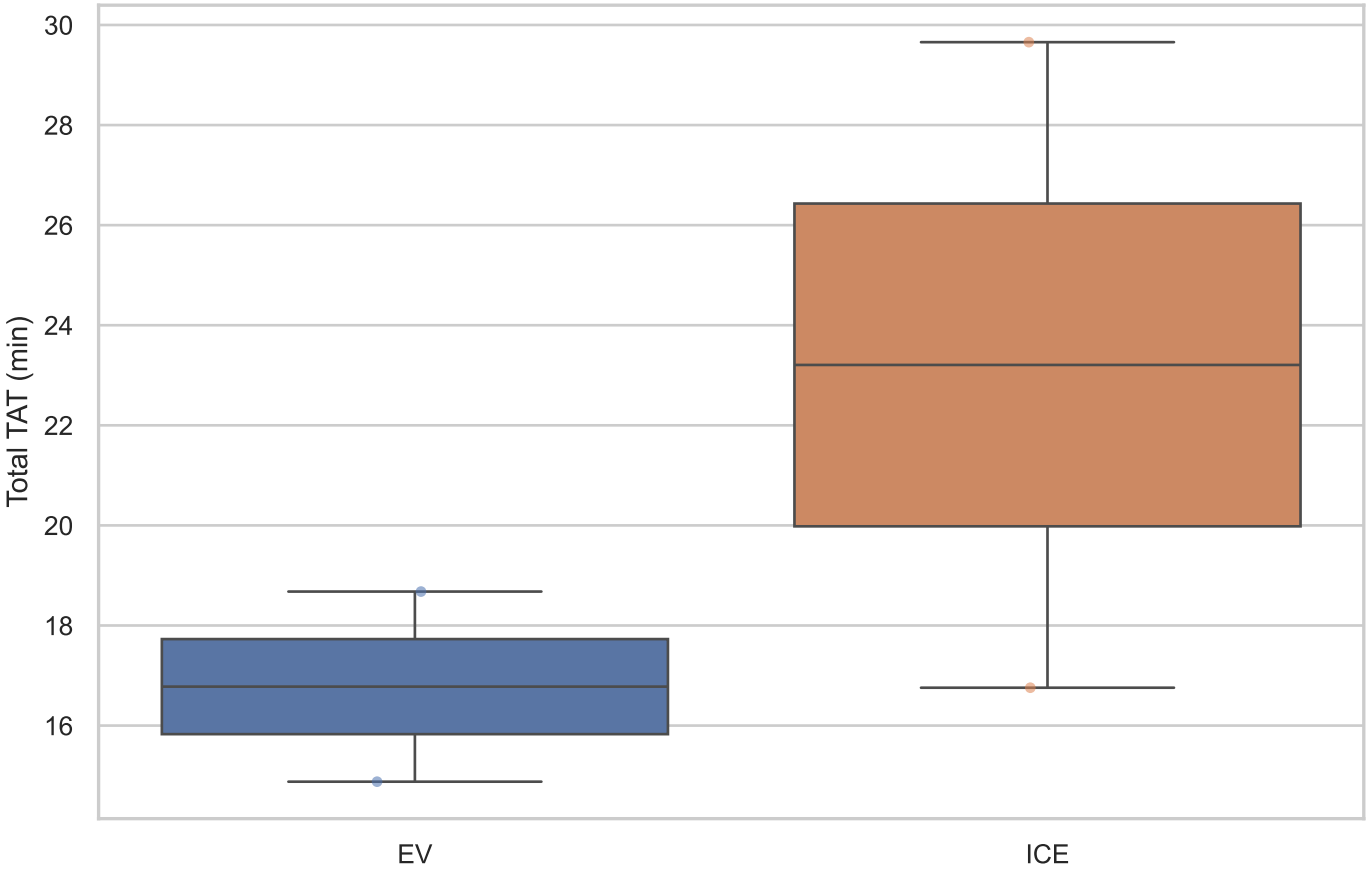
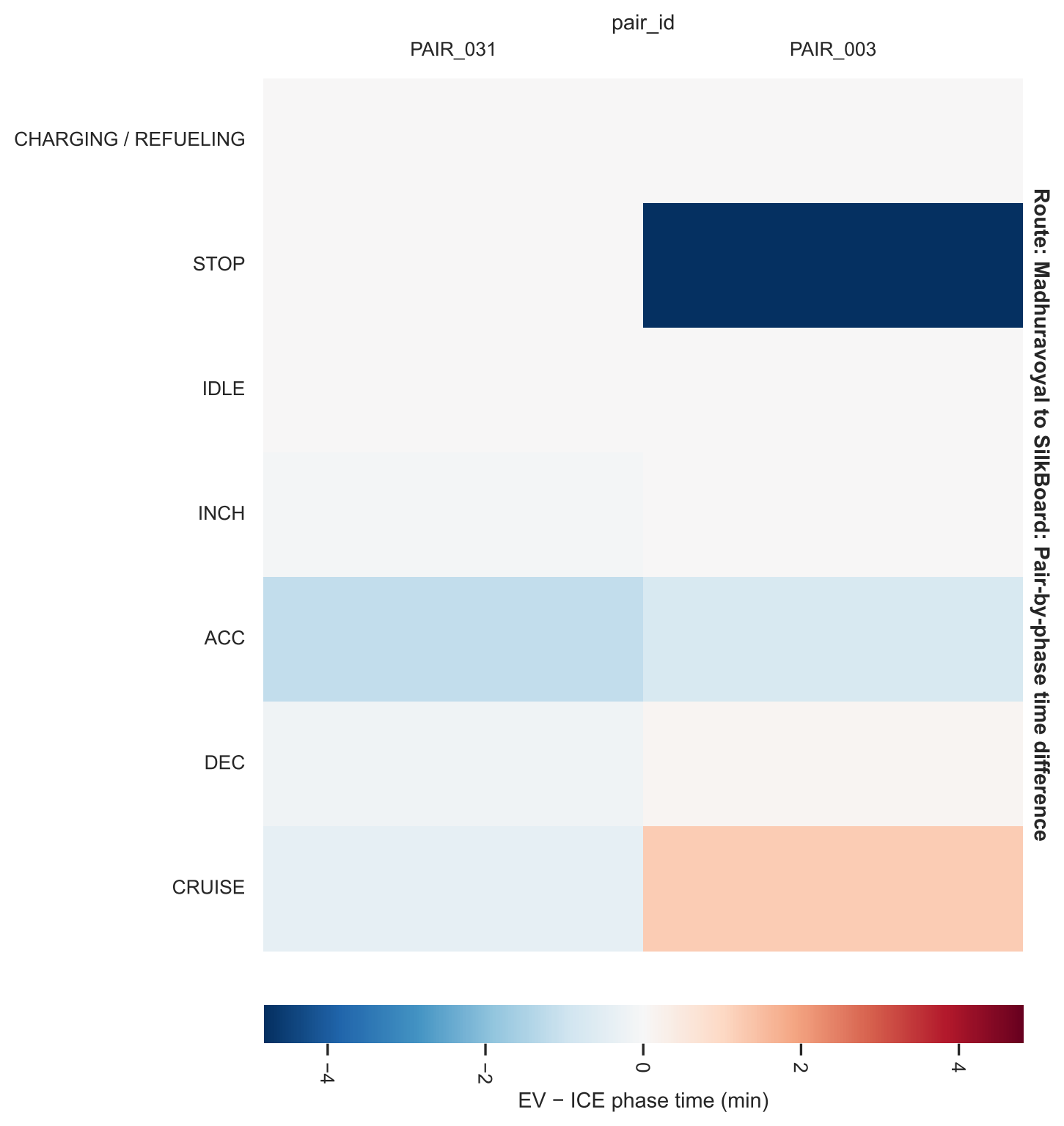
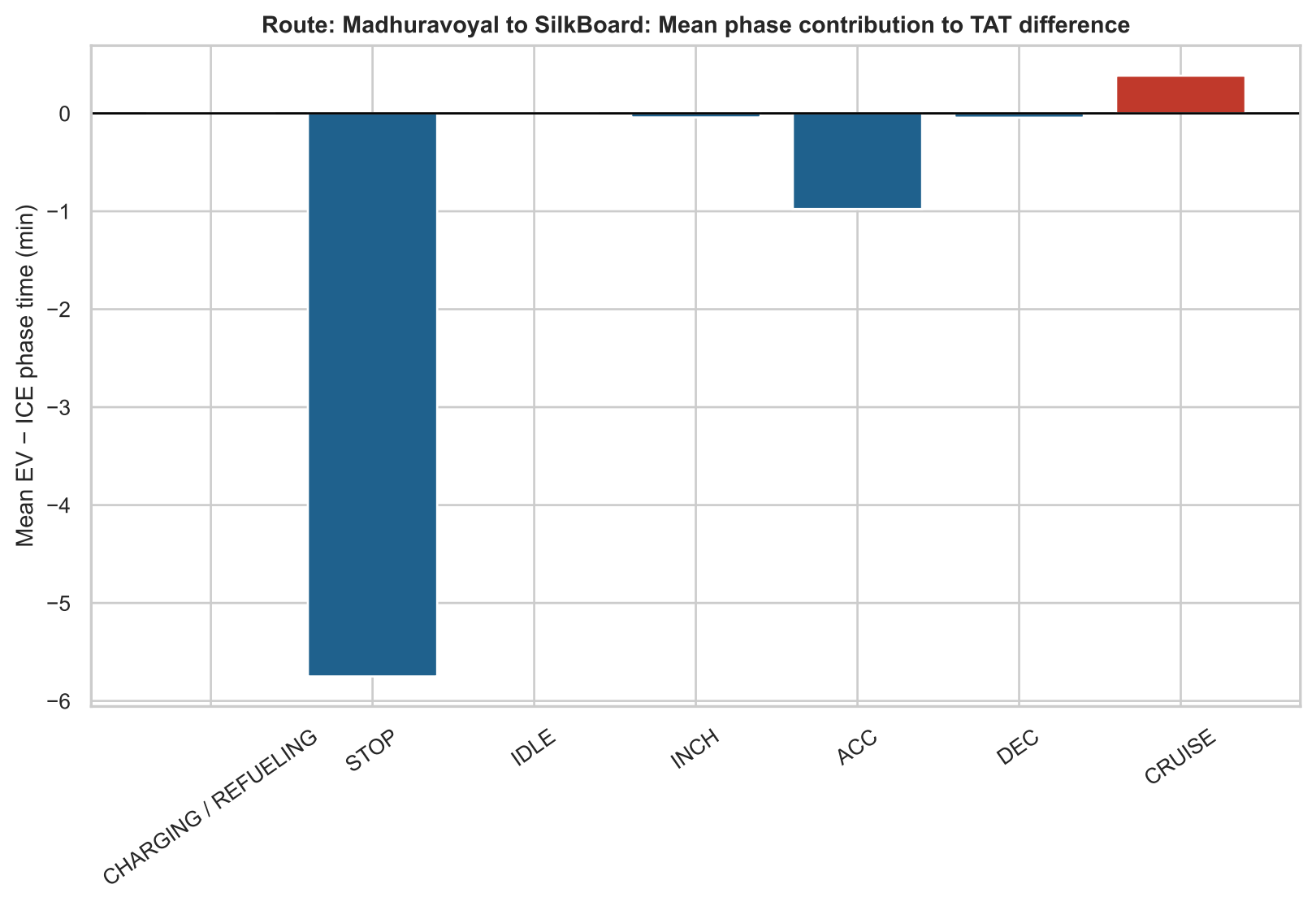
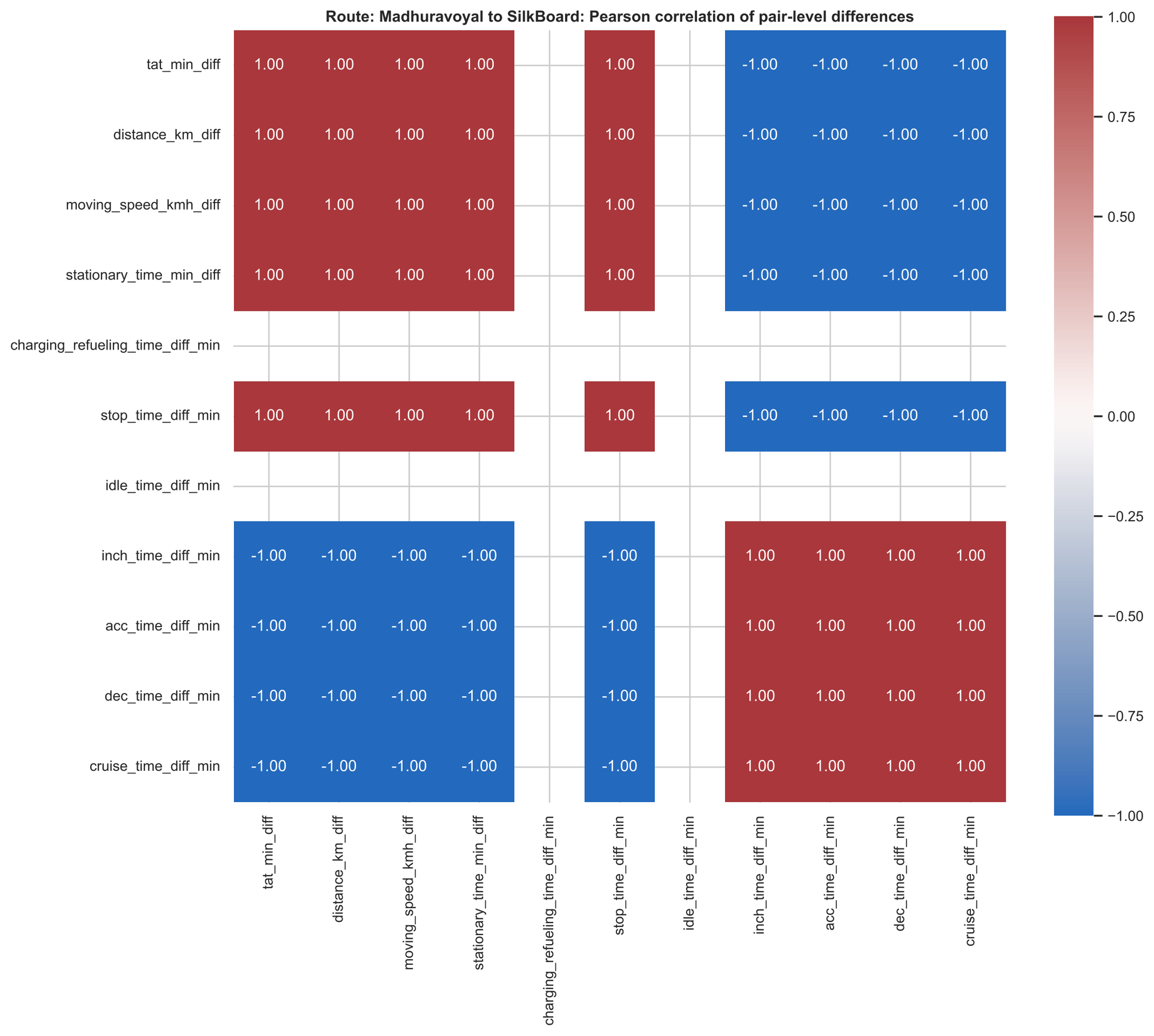

Route: Madhuravoyal to SilkBoard: Ranked pair-level TAT difference
PAIR_031
PAIR_003
−10
−8
−6
−4
−2
0
EV TAT − ICE TAT (min)

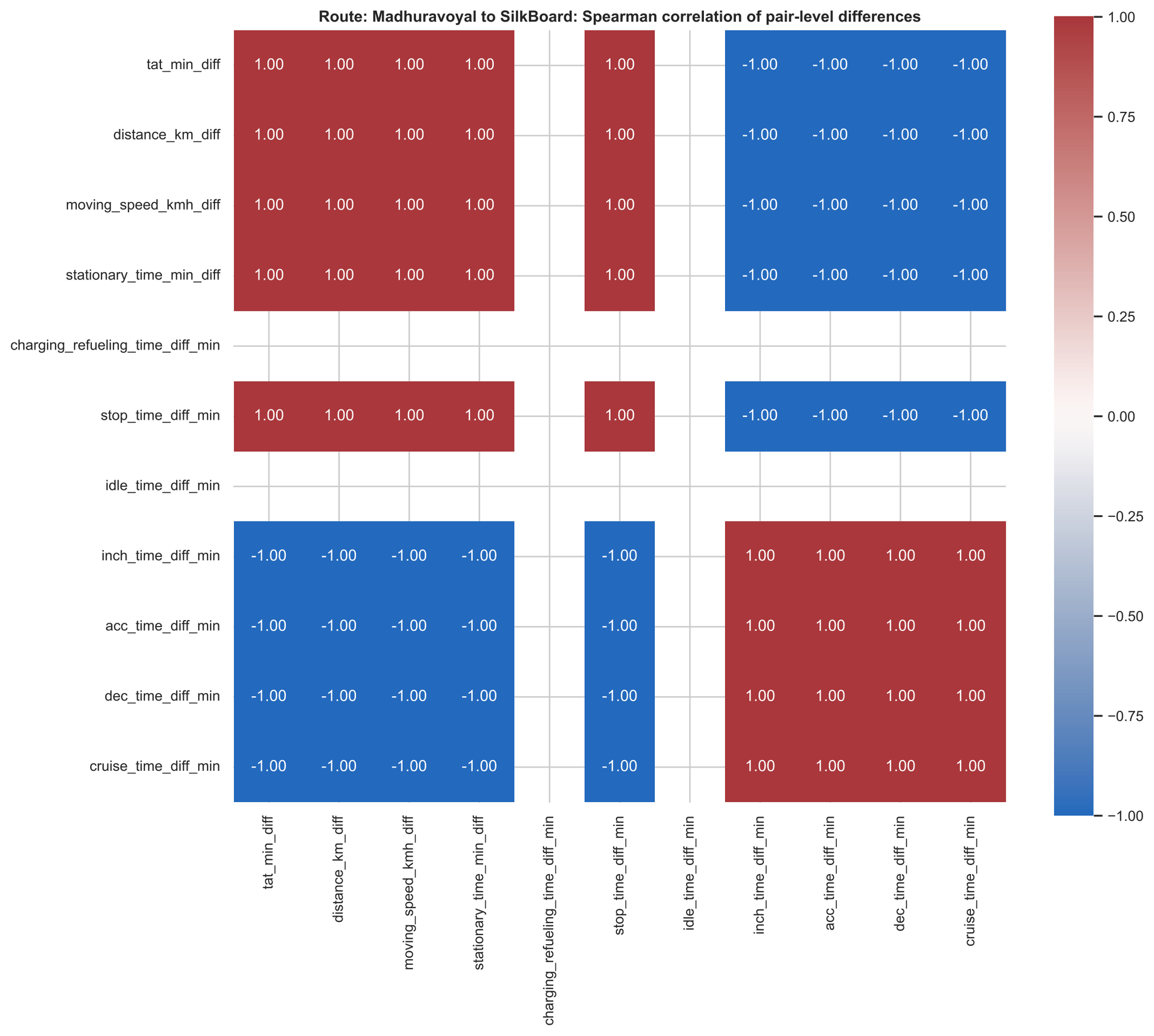

Route: Madhuravoyal to SilkBoard: Distribution of TAT differences
Count
1.0
0.8
0.6
0.4
0.2
0.0
−10
−8
−6
−4
−2
0
EV TAT − ICE TAT (min)

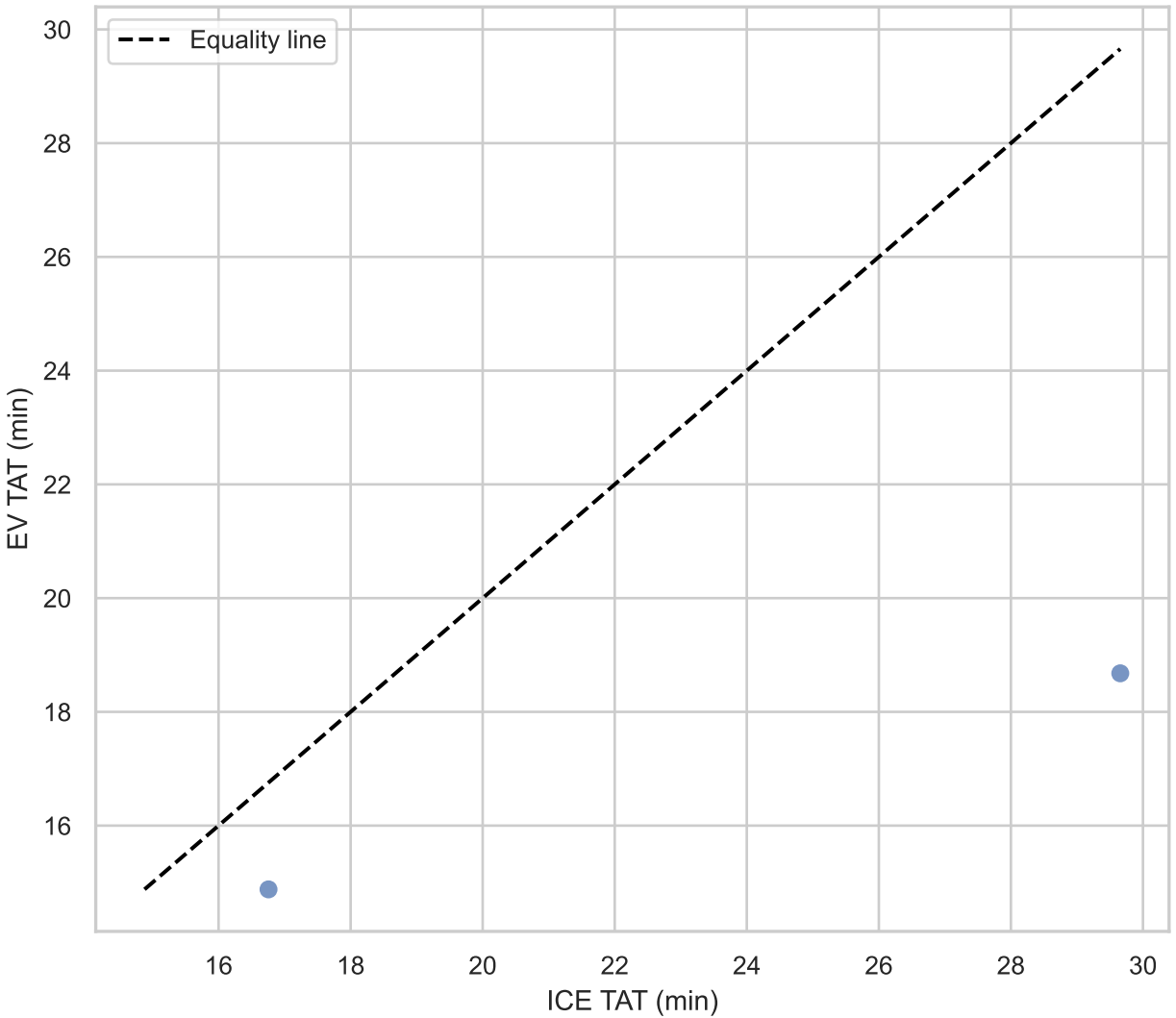

Route: Madhuravoyal to SilkBoard: EV and ICE TAT distribution
Total TAT (min)
30
28
26
24
22
20
18
16
EV
ICE

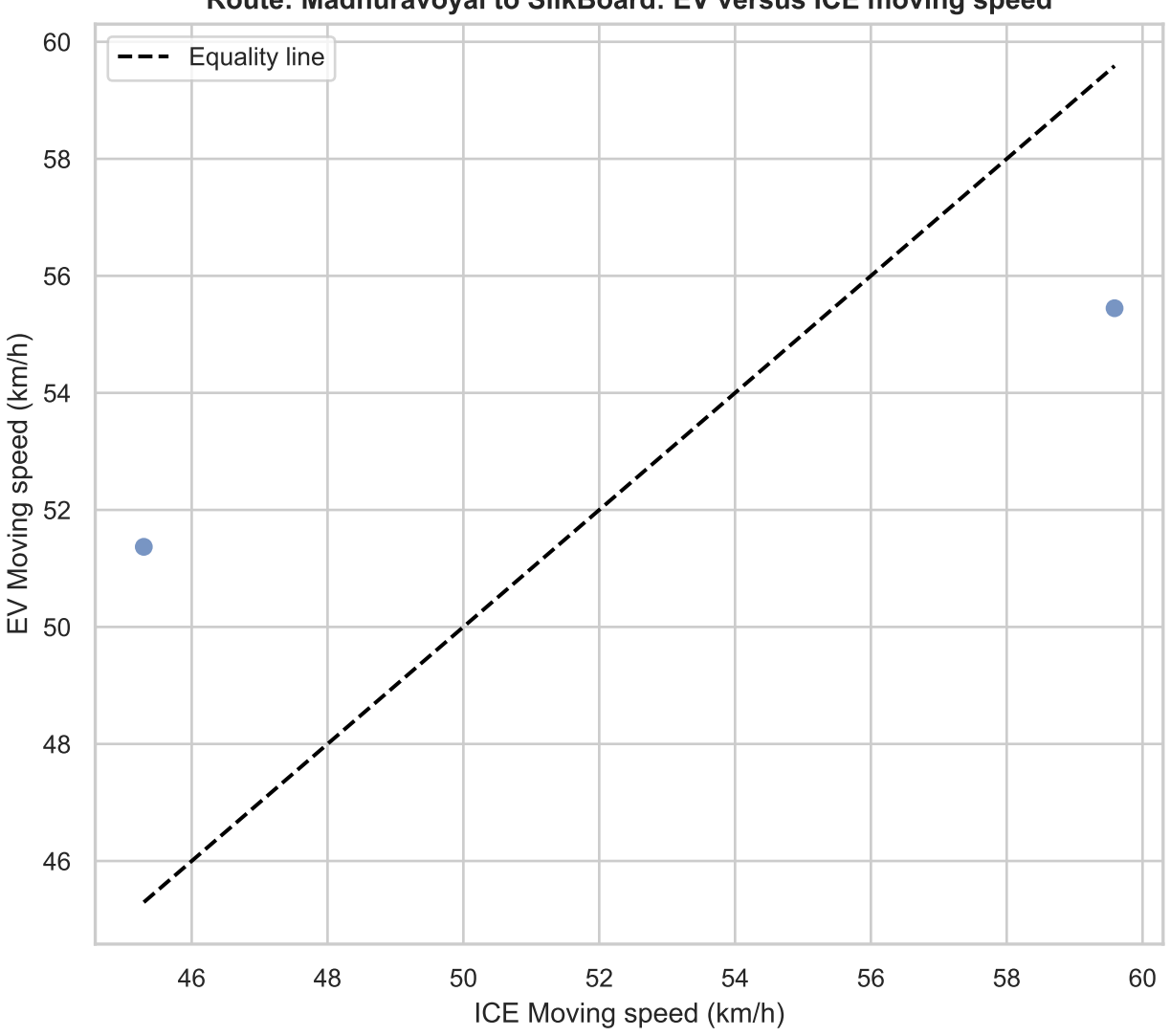

pair_id
PAIR_031
PAIR_003
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
Route: Madhuravoyal to SilkBoard: Pair-by-phase time difference
−4
−2
0
2
4
EV − ICE phase time (min)

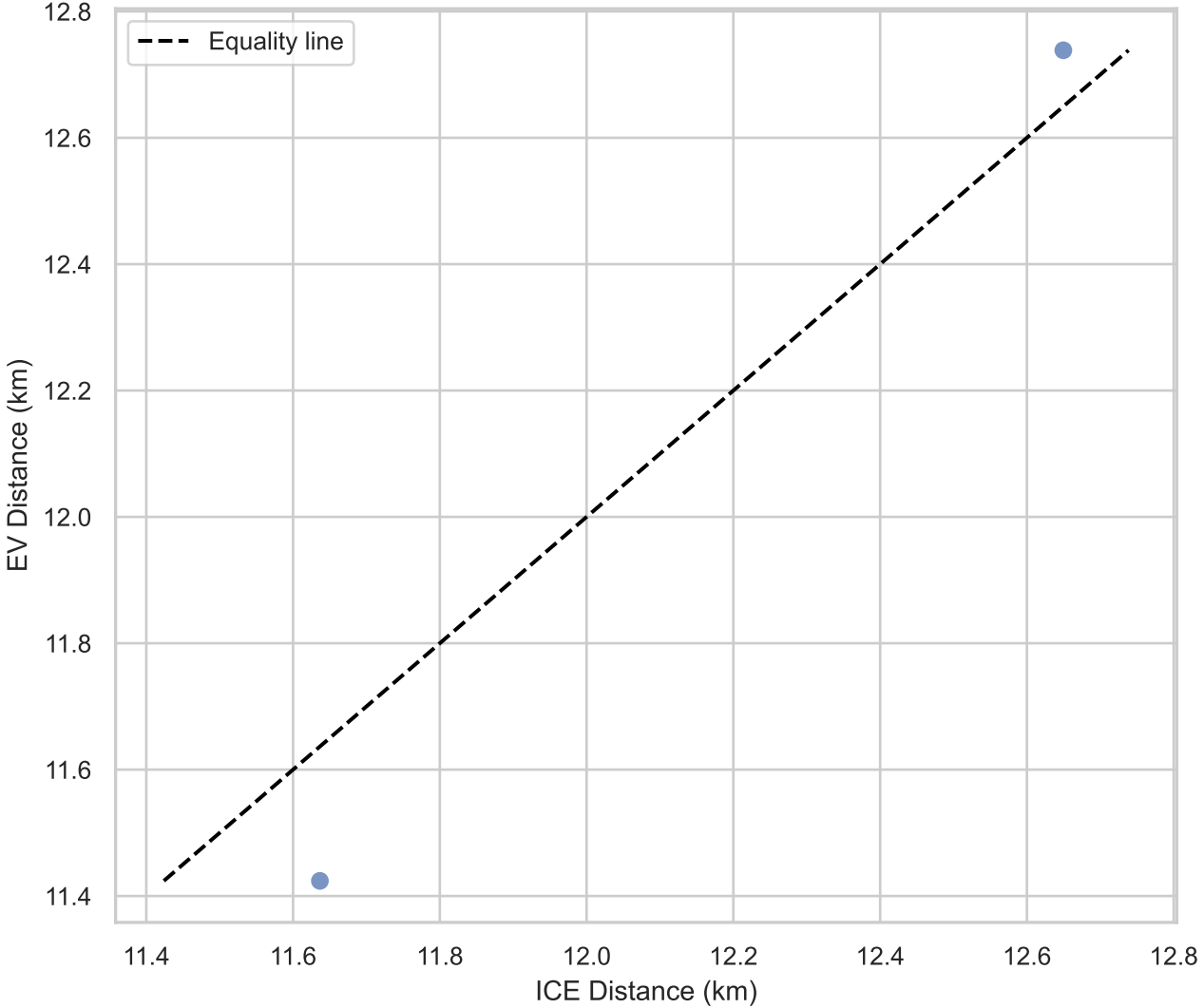

Route: Madhuravoyal to SilkBoard: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
0
−1
−2
−3
−4
−5
−6
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

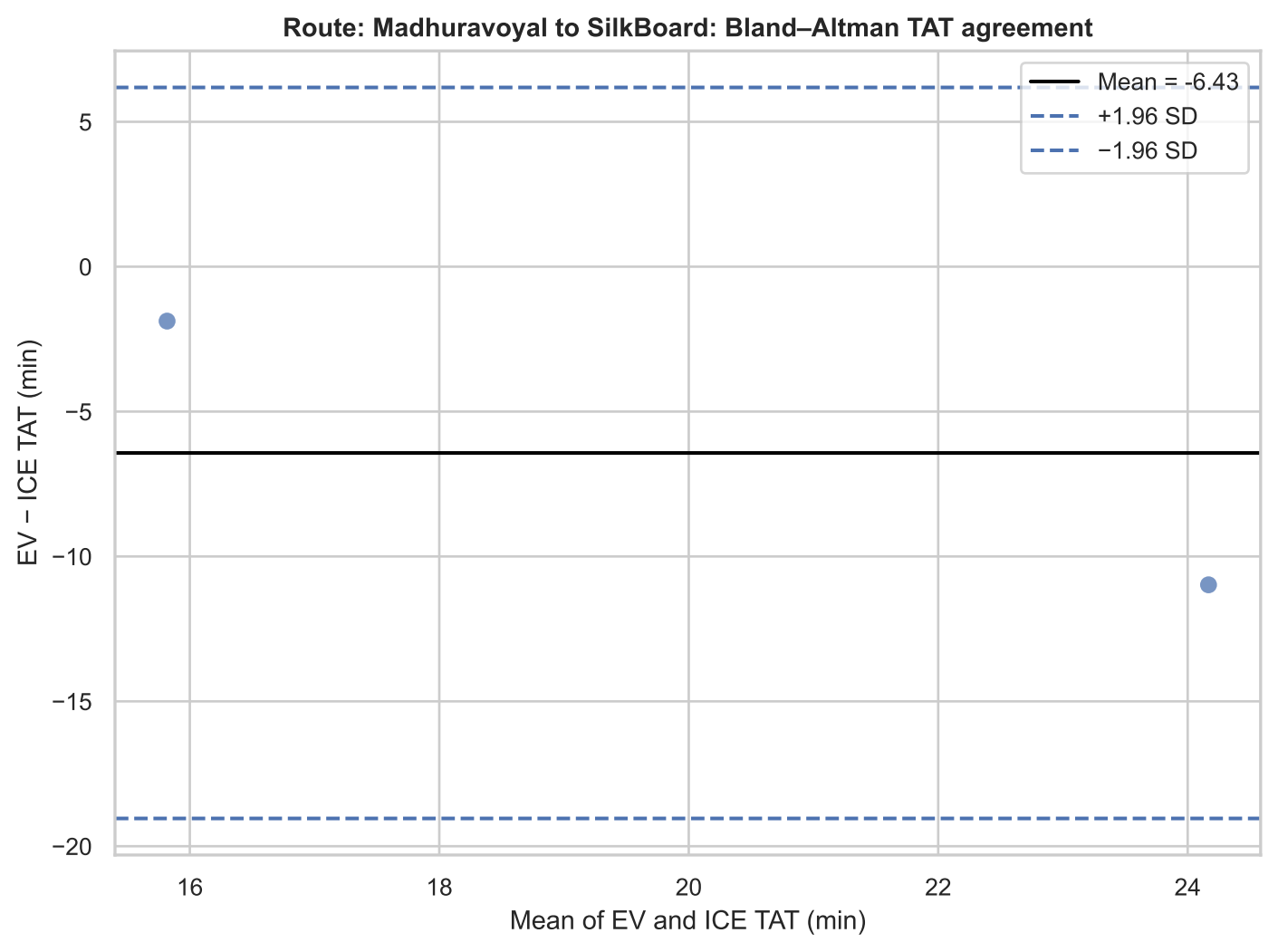

Route: Madhuravoyal to SilkBoard: Pearson correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

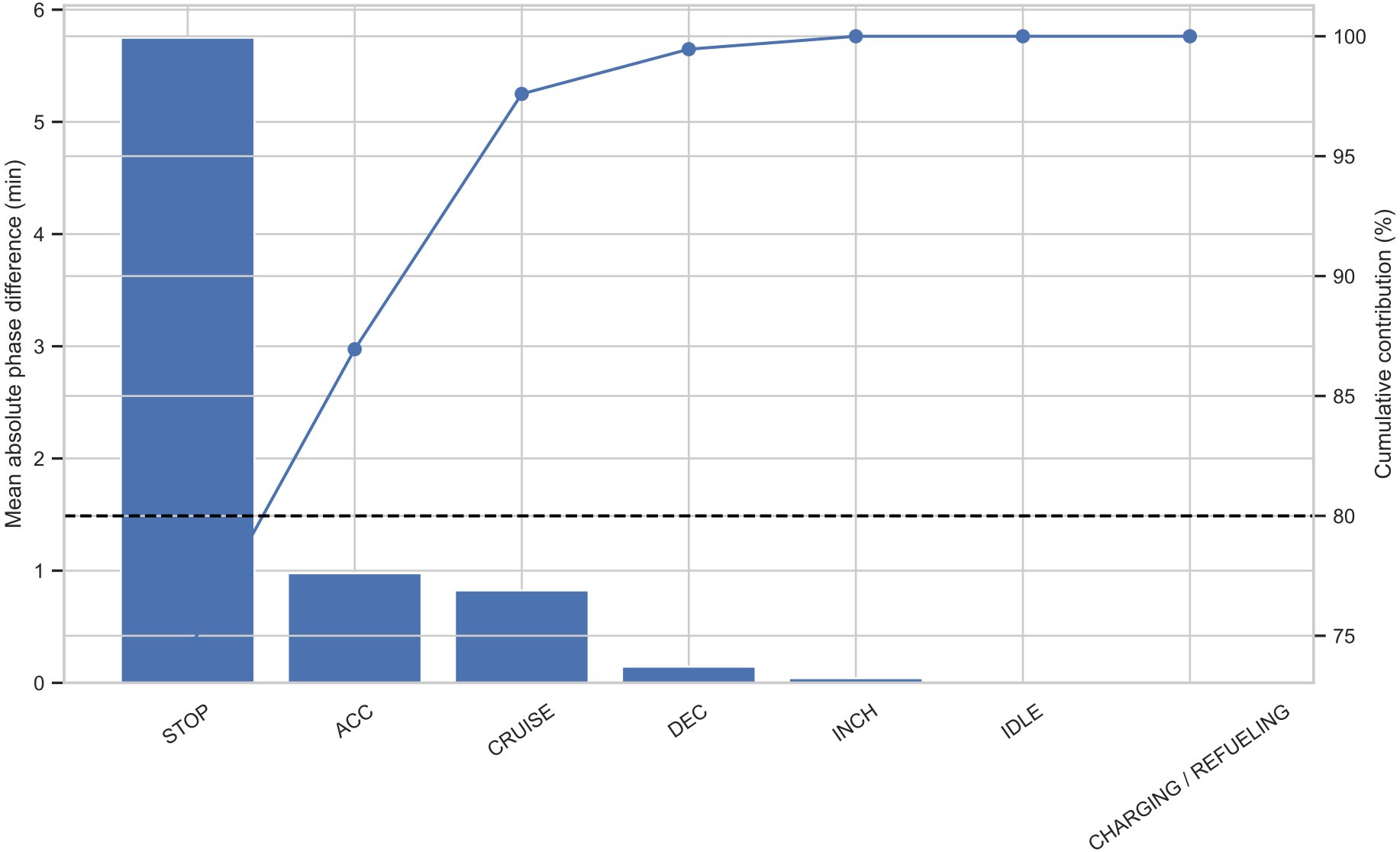

Route: Madhuravoyal to SilkBoard: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

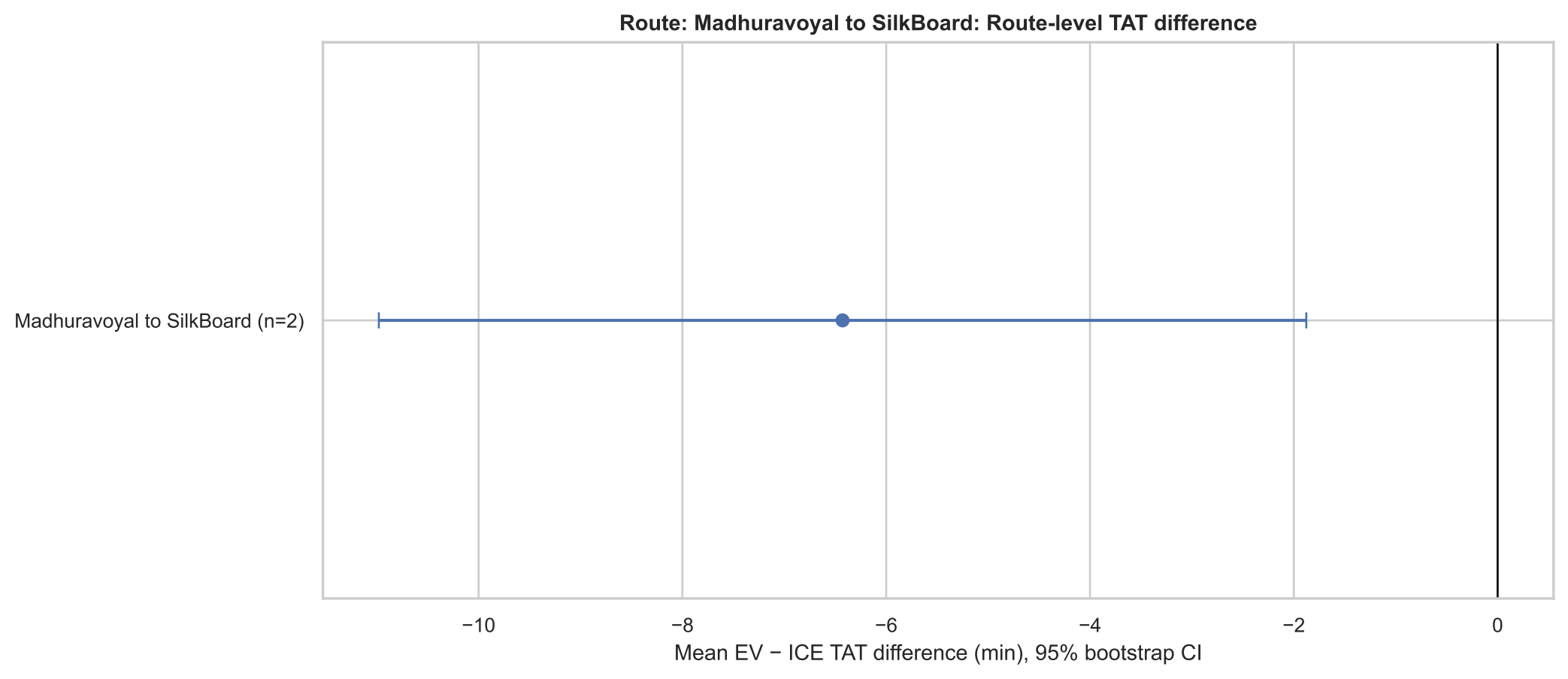

Route: Madhuravoyal to SilkBoard: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
30
28
26
24
22
20
18
16
16
18
20
22
24
26
28
30

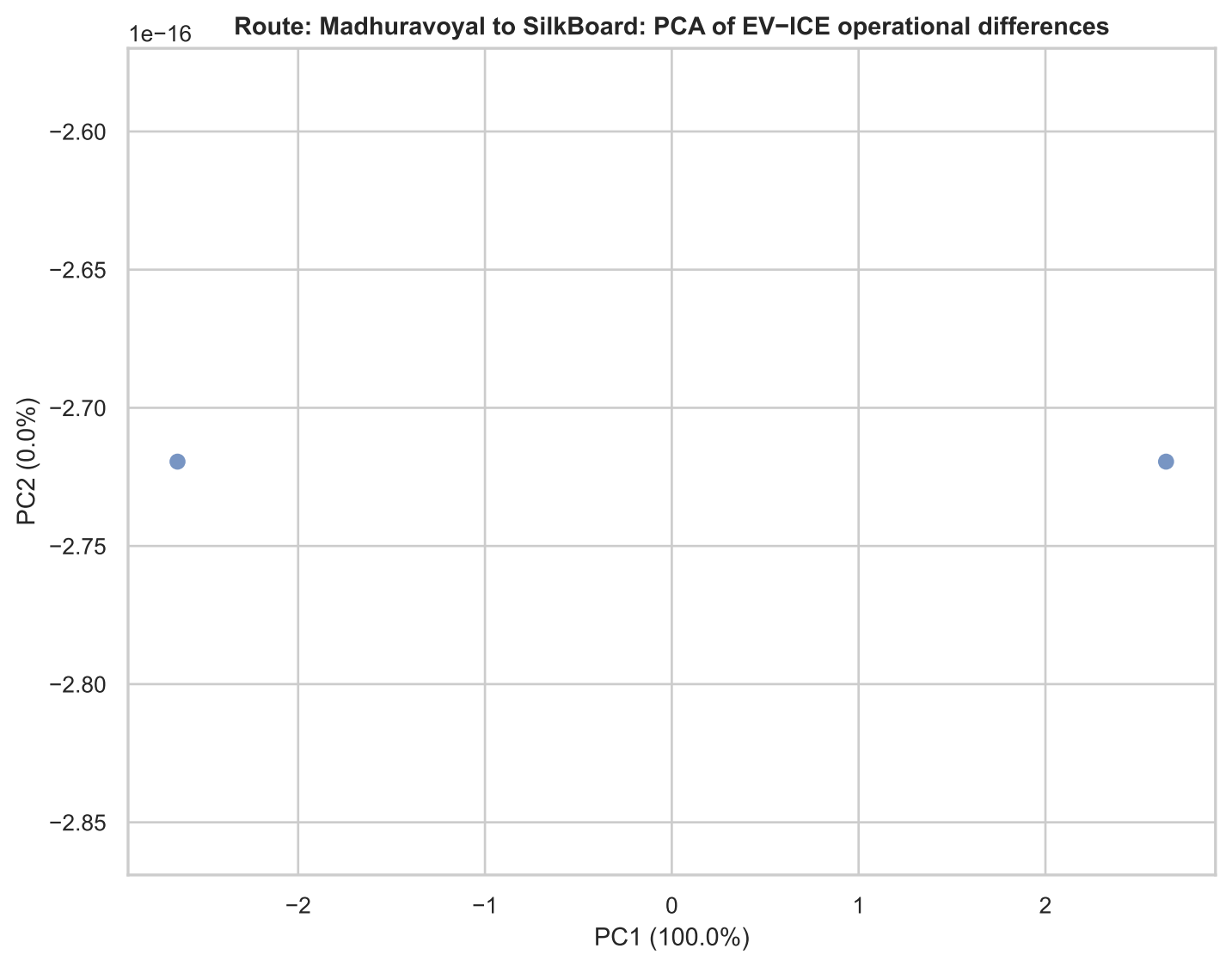

Route: Madhuravoyal to SilkBoard: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
60
58
56
54
52
50
48
46
46
48
50
52
54
56
58
60

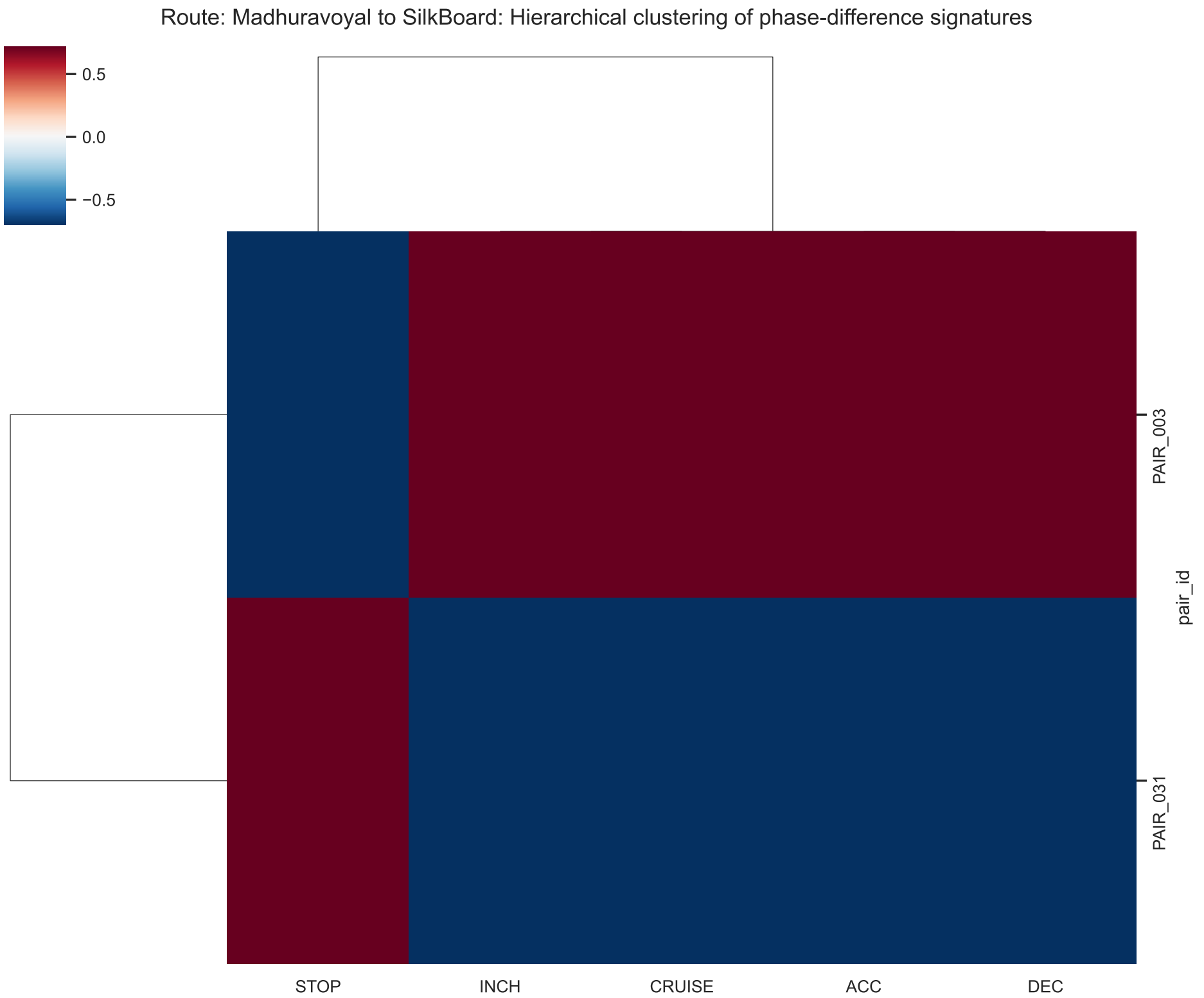

Route: Madhuravoyal to SilkBoard: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
12.8
12.6
12.4
12.2
12.0
11.8
11.6
11.4
11.4
11.6
11.8
12.0
12.2
12.4
12.6
12.8


Route: Madhuravoyal to SilkBoard: Bland–Altman TAT agreement
Mean = -6.43
+1.96 SD
−1.96 SD
5
0
−5
−10
−15
−20
EV − ICE TAT (min)
16
18
20
22
24
Mean of EV and ICE TAT (min)


Route: Madhuravoyal to SilkBoard: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
1
2
3
4
5
6
75
80
85
90
95
100
STOP
ACC
CRUISE
DEC
INCH
IDLE
CHARGING / REFUELING


Route: Madhuravoyal to SilkBoard: Route-level TAT difference
Madhuravoyal to SilkBoard (n=2)
−10
−8
−6
−4
−2
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Madhuravoyal to SilkBoard: PCA of EV−ICE operational differences
1e−16
−2.60
−2.65
−2.70
−2.75
−2.80
−2.85
PC2 (0.0%)
−2
−1
0
1
2
PC1 (100.0%)


Route: Madhuravoyal to SilkBoard: Hierarchical clustering of phase-difference signatures
0.5
0.0
−0.5
PAIR_003
PAIR_031
pair_id
STOP
INCH
CRUISE
ACC
DEC